



PORTABLE PRIVATE REPAYMENT HISTORY

Prove the record you earned at one
lender to **a lender who will never
see it.**

BUILT ON MIDNIGHT

COMPACT 0.23

APACHE-2.0



REPAY FOR THREE YEARS. WALK IN A STRANGER.

6

Licensed microfinance companies in Tunisia

ACM REGISTER

502,523

Active clients at Enda Tamweel alone

2024 • 80 BRANCHES • 24 GOVERNORATES

1M+

Tunisians served by Enda across its lifetime

CUMULATIVE, NOT ACTIVE



Enda Tamweel



Taysir Microfinance



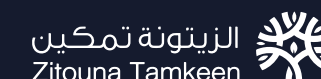
Baobab Tunisie



CFE Tunisie • DAAM



Advans Tunisie



Zitouna Tamkeen

Six institutions, overlapping territories. A good record stays locked inside the one that watched you build it. Borrow from any of the other five and the file starts **empty**.

THE DATABASE WOULD TAKE AN AFTERNOON. **NOBODY WILL BUILD IT.**

Reliable repayers are the book.

Sharing the list means handing every competitor your best customers. Every institution knows it, so **nobody shares**, and no amount of engineering changes that.

DEBT TRAVELS IN A DAY. GOOD BEHAVIOUR TRAVELS ONCE A MONTH.

Real time

That a loan exists

PERSON • CONTRACT • CLOSURE • WRITE-OFF

Monthly + 10 days

That it is being repaid on time

BALANCES • SCHEDULES • SCORE CHANGES

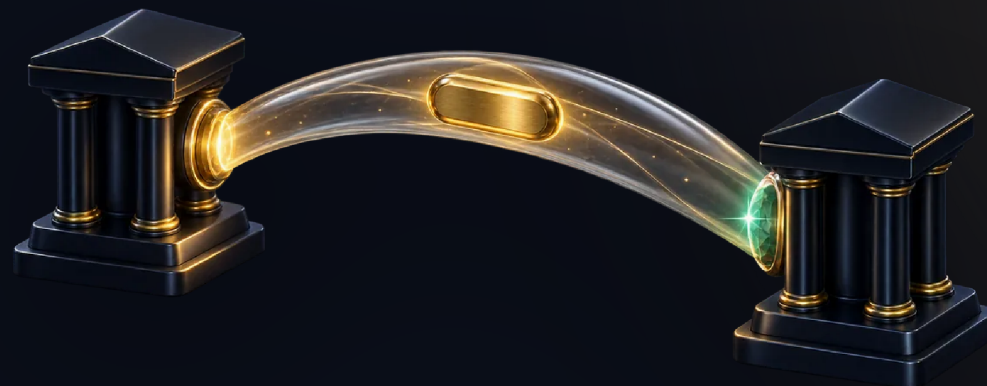
Recent good behaviour can be **up to forty days stale** at the moment the next lender decides.

A PROOF, NOT A FILE.

A The issuer publishes **one hash**. Nothing else.

B The borrower **holds** the credential.

C The verifier learns one fact: **cleared**.



8 + 6 CLEARS 14.

ON THE BORROWER'S DEVICE

Lender A	8 / 8
Lender B	6 / 7
Real total	14

ON THE MIDNIGHT LEDGER

Result	cleared ✓
Asked	≥ 14
Nullifier	0x4f2a...9c31



Not the total. Not which lenders. Not how many. **Forty repayments would look exactly the same.**

THE PRODUCT IS THE SPLIT BETWEEN TWO LEDGERS.

PUBLIC LEDGER

Admitted lender keys

Commitments and the current root

Thresholds and windows

Results and nullifiers

PRIVATE WITNESSES

The borrower's secret

Per-lender pseudonyms

Counts and reporting intervals

Which lenders contributed

UNLINKABLE IN PUBLIC. AGGREGABLE IN PRIVATE.

Per-lender pseudonyms

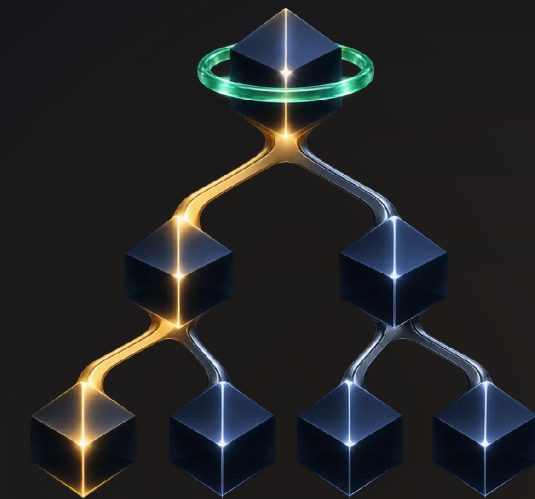
Two institutions cannot tell they share a customer. The circuit still adds across them.

Revocation by leaf overwrite

A plain MerkleTree, deliberately not Historic. No revoked-list, so nothing leaks.

Uniform proof shape

Four root checks every time. How many credentials you hold never leaks.



SIX CIRCUITS. SIXTY-THREE TESTS. ZERO MOCKS.

6

Compact circuits

COMPILER 0.31.1

40

Circuit tests

AGAINST COMPILED CONTRACTS

23

API integration tests

FULL LIFECYCLE



Transcript equality

1 RECORD $\equiv$ 4 RECORDS

One test runs the circuit with one credential and with four, and asserts the **public transcripts are byte-identical**. The privacy claim is a test that fails if we break it.

START WHERE TWO INSTITUTIONS IS ENOUGH.

BEACHHEAD

Any two of Tunisia's six licensed MFIs. A positive-evidence supplement, not a replacement.

THEN

MFI to bank graduation. Same proof, higher threshold.

THEN

Francophone Africa and MENA. Same market structure.

PRIMITIVE

Anywhere competitors must trust each other's records.

The verifier pays. Issuing is free. Borrowers never pay.

WHAT IT DOES NOT DO YET.

LIMITS TODAY

Issuer trust is the root · ZK cannot prove a repayment happened

1,024 leaves, four lenders per proof

Unencrypted browser storage

No trusted clock on chain

WAVES 2 AND 3

Encrypted storage and recovery

Larger tree, issuance batching

Minimum distinct-lender proofs

Issuer governance, lender removal



A BORROWER CARRIES A PROOF, NOT A FILE.

[GITHUB.COM/FAROUK-ALLANI/AMANA](https://github.com/farouk-allani/amana)

APACHE-2.0

MIDNIGHTNTWRK

